

Chenyu Wang

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LLMs for Software Engineering • Coding Agents • AI Security & Program Analysis

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PhD candidate and Research Engineer at SMU's SOAR Lab (advisor Prof. David Lo), building and securing LLM-based coding systems: inference-time control of coding agents, robustness of code models, and repository-scale vulnerability detection. First-author research at **ASE** and **EASE** (invited to *IEEE Software*), plus a paper under review at a top-tier (**CORE A***) venue; co-authored work at **ICSE**, **TSE**, and **MSR**.

Experience

Singapore Management University – SOAR Lab 2023 – Present
Research Engineer • Advisor: Prof. David Lo Singapore

- Lead first-author across the lab's **inference-time agent control** and **code-model security** line (under review at a CORE A* venue; ASE 2025; EASE 2026 → *IEEE Software*), owning each project **end-to-end** – from empirical study and dataset construction to model training, evaluation, and deployable tooling.
- With GovTech Singapore** (since Jul 2026): building a **repository-level, agentic vulnerability-detection** pipeline – **code property graphs** for inter-procedural analysis, plus an **LLM-as-a-judge** stage that prunes candidate vulnerable paths to focus an expensive detection loop, under real CI/CD cost constraints.

Education

Singapore Management University – Ph.D., Computer Science (part-time) 2023 – 2027 (expected)
Software Engineering & Trustworthy AI • Advisor: Prof. David Lo Singapore

Singapore University of Technology & Design – B.Eng., Computer Science & Design May 2019 – Aug 2022
GPA 4.83/5.0, Highest Distinction • Minor in Artificial Intelligence • NUS exchange (2021) Singapore

Selected Publications (first author)

Fail-Fast, Restart-Smart: Early Failure Prediction and Restart for SWE Agentic Tasks Under review, 2027 **CORE A***

A **0.6B** monitor supervises **27B+** coding agents *at inference*: **14.6-20.4% execution-token savings** at 5% false-positive rate and issue resolution **66.6% → 71.8%** on SWE-bench Verified; one monitor **transfers across 4 agents**, including a closed-API model – directly cutting agent cost in production. arXiv:2608.03222

Backdoors in Code Summarizers: How Bad Is It? ASE 2025 **CORE A***

First **systematic (9-factor)** study of data-poisoning backdoors in code LLMs: just **20 of 454,451 samples (0.004%)** implant a backdoor the standard spectral-signature defense **fails to remove** – **25×** below rates the field assumed safe. Exposes a concrete AI supply-chain risk for code models.

Industry Practitioners' Perspectives on AI Model Quality EASE 2026 • invited to *IEEE Software* **CORE A**

Mixed-methods study (**15 interviews + 50-practitioner survey**) mapping how industry actually prioritizes AI quality: **efficiency is treated as a hard constraint over correctness** in real-time systems – a design contract for cost-aware ML.

Selected Publications (co-authored)

- Unveiling Memorization in Code Models ICSE 2024 **CORE A***
- Gotcha! This Model Uses My Code! Evaluating Membership Leakage Risks in Code Models IEEE TSE 2024 **Top journal**
- What Do Users Ask in Open-Source AI Repositories? An Empirical Study of GitHub Issues MSR 2023 **CORE A**
- Are We Ready to Embrace Generative AI for Software Q&A? ASE 2023 **CORE A***

Awards & Recognition

- Best Student Award**, SUTD – ranked **1st of 400+ students** (Term 3) • Honours with Highest Distinction • Honours List (multiple terms).
- Senior Middle 2 Scholarship**, Singapore Ministry of Education • Selected for **NUS exchange** via SUSEP (7 students from pillar).
- Reviewer**: ACM TOSEM (×2), MSR 2026 • **Invited extension** to *IEEE Software* (EASE 2026 study).

Technical Skills

Languages: Python, Java, JavaScript (*proficient*); C, TypeScript, Shell (*working*)

ML / LLM: PyTorch, Hugging Face, LoRA / PEFT fine-tuning, LLM agents, reward & monitor models, code LLMs (CodeT5, PLBART, Qwen, Gemma)

Program Analysis & Security: code property graphs, program slicing, taint analysis, backdoor & adversarial robustness, vulnerability detection

Engineering: React, Angular, Flask, Firebase, Docker, Git, Linux • empirical evaluation & LLM-agent benchmarking (SWE-bench)